

VOX-OMEGA

Final Verified 60-Day Build Plan

Emotional, low-latency, code-switched voice AI - clean-room, open-stack, production-first

Output	Downloadable PDF build plan with source-backed research, day-by-day execution, repo spec, metrics, and safety guardrails
Primary constraint	2 months of vacation - fastest feasible path, no from-scratch pretraining
Recommended path	Moshi-style speech-to-speech + Mimi codec + emotion-conditioned TTS + Hinglish data engine + DPO/GRPO alignment
Reality check	A public demo can be shipped in 60 days; a true frontier-scale Hume/Silk competitor requires more data, compute, and team time
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This plan intentionally avoids copying proprietary systems, private weights, or unconsented voices. The correct strategy is a clean-room build that learns from public research and your own consented datasets.

Prepared from: user-provided VOX-OMEGA materials, Vatsal Bharti resume screenshot, public research papers, and current web sources available during generation.

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1. Immediate answer - why the PDF did not appear earlier

You asked for a final PDF. The previous reply stopped before producing the file. This document is the actual downloadable output. I rebuilt the plan into a fresh, verified PDF instead of only pointing at the earlier uploaded bundle.

The earlier uploaded VOX-OMEGA master plan is useful and detailed, but it mixes a 6-month ambitious path with a 2-month vacation constraint. This final version resolves that conflict: the 60-day path uses existing open models, clean-room fine-tuning, strong data engineering, and a production streaming wrapper. From-scratch frontier pretraining is explicitly excluded from the 60-day plan.

2. Executive decision - the only viable 60-day strategy

The fastest credible route is not to train a new speech foundation model from scratch. The fastest route is to assemble and fine-tune an open speech-to-speech stack, then differentiate on data, emotion control, latency engineering, and product demo quality.

Decision	Final call	Why
Base system	Fork Moshi-style full-duplex speech-to-speech	Moshi is a public real-time full-duplex speech framework with reported 160 ms theoretical and 200 ms practical latency [R1].
Codec	Use/fine-tune Mimi-style codec, do not retrain codec first	Codec pretraining burns weeks. Freeze or lightly adapt first; revisit codec only after MVP.
Emotion	Inline tags + continuous V/A/D controls	EmoVoice shows LLM-based TTS can use freestyle natural-language emotion control [R2].
Code-switch	Build Hinglish/Tanglish data engine	Code-mixed Hinglish has spelling inconsistency and data scarcity, so synthetic + curated real data is the moat [R9].
Voice cloning	Consent-only 10-30 sec reference mode	OpenVoice and Hume-style custom voices show short-clip cloning is feasible, but misuse risk is high [R3][R5].
Serving	WebRTC + low-jitter streaming + vLLM-style KV cache ideas	PagedAttention/vLLM improves serving throughput through KV cache paging and batching [R4].
Safety	Watermark + consent phrase + abuse monitoring	Audio watermarking is a minimum requirement for voice-cloning products [R7][R10].

Minimum Day-60 win condition: a live web demo with 3-5 expressive Hinglish personas, interruptible speech, average first audible response below 300 ms on a GPU server, and human A/B preference over your baseline stack on emotional appropriateness.

3. Verification log - what is sourced vs. inferred

Claim	Status	Evidence
Moshi is built to avoid traditional VAD/ASR/LLM/TTS pipeline latency and supports full-duplex dynamics	Sourced	Moshi paper abstract: pipeline systems have seconds of latency and lose non-linguistic information; Moshi models speech-to-speech with parallel user/model streams [R1].
Moshi reports 160 ms theoretical, 200 ms practical latency	Sourced	Moshi paper abstract [R1].
Freestyle natural-language emotion control is an active TTS approach	Sourced	EmoVoice paper describes LLM-based emotional TTS with fine-grained freestyle prompting [R2].
Short-clip voice cloning is plausible	Sourced	OpenVoice paper requires only a short audio clip; Hume EVI 3 reporting says 30-second custom voice samples [R3][R5].
Hinglish data is hard due to inconsistent spelling and limited high-quality data	Sourced	Hinglish conversational AI paper [R9].
Silk likely uses a CSM/Mimi-like recipe	Inferred	Based on your uploaded notes and public architecture similarities. Treat as hypothesis, not verified source truth.
<80 ms end-to-end TTFB is Day-60 achievable	Rejected for solo 60-day plan	Use <300 ms practical target first. <80 ms can be a later optimization milestone requiring deep kernel/server work.
Beat Hume/Silk globally in 60 days	Rejected as guaranteed claim	A demo can outperform on a narrow Hindi-English emotional use case; global superiority requires independent human eval and scale.

This PDF uses the words 'beat' and 'surpass' as measurable engineering targets, not marketing fantasy. Every target below has an evaluation gate.

4. Competitor teardown

4.1 Hume EVI 3 - what matters

- Hume's differentiated angle is affect: recognizing and generating emotionally nuanced voices, not just high-quality TTS.
- Reporting on EVI 3 says it can recreate voices from a 30-second clip and currently focuses on English and Spanish [R5].
- Recent reporting says Google DeepMind hired Hume's CEO and several engineers through a licensing/talent arrangement, reinforcing that emotional voice is strategically valuable [R6].

4.2 Rumik Silk - what to emulate without copying

- Your uploaded plan characterizes Silk as code-switched, emotional Indic voice AI, with Hinglish/Tanglish/Manglish/Kanglish as core use cases.
- The clean-room path is to reproduce the category: multilingual code-switch voice generation, inline emotion control, low latency, and companion-like persona handling.
- Do not scrape private datasets, reverse-engineer deployed endpoints, clone proprietary voices, or copy model outputs at scale. Build your own data flywheel.

4.3 Moshi - why it is the fast path

Moshi directly attacks the latency and turn-taking problem by moving from a pipeline of separate VAD/ASR/dialog/TTS components to a speech-to-speech model with parallel streams. This is the most important architectural shortcut for a 60-day build [R1].

4.4 CSM/Mimi-style TTS - where it fits

A CSM/Mimi-style TTS path is excellent for high-quality speech generation, but in the strict 60-day plan it should be used for fine-tuned TTS/persona generation, not as a from-scratch foundation model project.

4.5 OpenVoice and short-reference cloning

OpenVoice provides a public reference point for short-clip, cross-lingual, flexible-style voice cloning; it supports emotion, accent, rhythm, pauses, and intonation control beyond simply copying the reference voice [R3].

5. Product target - what must be true by Day 60

Dimension	Day-60 target	Stretch target
First audible response	<300 ms p50 after warm cache	<200 ms p50
Full-duplex/interruption	Stop or fade output within 250 ms of user interruption	<180 ms
Emotion control	10 core emotions + intensity tags + pauses/laughs/sighs	Continuous V/A/D control
Languages	English + Hinglish first	Hinglish + Tanglish + Telugu-English
Personas	3 strong personas	10 personas
Voice cloning	Consent-only same-user voice demo	30-sec gold-tier voice clone
Quality eval	Human A/B beats baseline on emotion by 60%+	Beats selected commercial demo clips in blind tests
Safety	Watermark, consent phrase, no public arbitrary cloning	Abuse dashboard + automated misuse flags

Do not chase a giant model. Chase a demo that sounds alive. Emotional appropriateness, turn timing, and Hinglish naturalness will beat parameter count in user perception.

6. Final architecture - 60-day VOX-OMEGA system

USER MIC 24 kHz
-> WebRTC ingest

- > streaming VAD + interruption monitor
- > Moshi-style speech-to-speech core OR low-latency STT fallback
- > dialog planner: small LLM creates reply + emotion plan
- > emotion-conditioned TTS core
 - text tokens + emotion tags + speaker/persona prefix
 - > codec tokens
 - > Mimi/codec decoder
- > output audio chunks over WebRTC

Control plane:

- persona memory -> prompt/prefix
- consent service -> blocks unauthorized voice cloning
- watermark service -> marks generated audio
- eval logger -> latency, WER, emotion, MOS, user rating

6.1 Architecture principle

Build a two-track system. Track A is the full-duplex speech-to-speech path for the best demo. Track B is a robust modular fallback: streaming STT + dialog LLM + streaming TTS. Track B keeps the product alive if fine-tuning Moshi internals takes longer than expected.

6.2 Model stack

Layer	Recommended implementation	Fallback
Speech core	Moshi-style open speech-to-speech model	STT + LLM + TTS pipeline
Codec	Mimi-compatible codec, frozen first	Existing codec from chosen base
TTS	Codec-token AR model with emotion tag SFT	OpenVoice/Fish-Speech/XTTS-style serving for prototype
Emotion	Tag tokens + V/A/D embedding + persona prompt	Tag-only SFT
Speaker	Reference encoder + consent gate	Fixed voices only
Serving	CUDA graphs, warm KV cache, WebRTC	FastAPI websocket

6.3 Emotion control grammar

```
<persona:swati_mumbai_warm>
<voice_ref:consented_user_001>
<emotion valence=0.7 arousal=0.8 dominance=0.35>
<excited intensity=0.85> arre yaar, seriously? <laugh soft dur=0.4>
<pause ms=180> that is actually insane.
```

Start with tags. Only add continuous FiLM-style modulation after tag control works. Emotion tags are easy to debug; continuous controls are powerful but hard to attribute.

7. Data moat - the actual place to win

7.1 Data tiers for the 60-day plan

Tier	Target by Day 60	Source strategy	Why it matters
T0 Baseline speech	20-50 hr	Open datasets + clean conversational clips	General pronunciation and codec stability
T1 Hinglish text	100k-500k lines	Synthetic + curated public corpora	Code-switch grammar and spelling variation
T2 Emotional audio	10-30 hr	Synthetic TTS + your own recordings	Tags/laughs/sighs/pause behavior
T3 Real studio micro-set	2-5 hr	You + friends/actors, consented	Human emotional cadence; synthetic alone is not enough
T4 Preference pairs	2k-10k pairs	Generate two samples, judge with human + model	DPO-lite alignment
T5 Safety data	500-2k prompts	Abuse cases, impersonation attempts	Prevents irresponsible launch

7.2 Synthetic Hinglish dialogue engine

```
for theme in themes:
    dialogue = LLM.generate(
        language_mix='Hinglish Latin script',
        emotions=['warm', 'excited', 'sad', 'playful', 'sarcastic'],
        tags=['<laugh>', '<sigh>', '<pause ms=200>'],
        turns=4
```

```

)
audio = base_tts.synthesize(dialogue, voice=random_consented_voice)
transcript = whisper_backtranscribe(audio)
keep if WER < 12% and emotion_classifier_consistency > threshold

```

7.3 Real recording script

Record 500-1000 short utterances across five emotions: warm, excited, sad, playful, frustrated. Use 24 kHz mono WAV, quiet room, fixed mic distance, and written consent. Real emotional micro-data will outperform large weak synthetic data for the last layer of perceived warmth.

7.4 Filtering rules

- Drop any sample with high background noise, clipping, or speech overlap unless it is explicitly for interruption training.
- Back-transcribe generated audio and reject high WER samples.
- Use an emotion classifier only as a filter, not as ground truth. Human spot checks decide final quality.
- Deduplicate speakers, scripts, and near-identical synthetic lines to prevent memorized cadences.

8. Training plan - what to fine-tune, what to freeze

8.1 60-day training sequence

Stage	Days	Action	GPU	Gate
0	1-3	Run base model inference and baseline latency eval	Local + rented GPU	Can talk end-to-end
1	4-10	Build data pipeline and 2 hr real emotional corpus	CPU + API	Dataset manifest passes QA
2	11-18	Tag-only LoRA SFT on emotion/Hinglish data	1x A100/H100 or 2x 4090	Emotion tags audible
3	19-26	Scale synthetic data to 20-50 hr and retrain	1x A100/H100	WER and MOS stable
4	27-35	Streaming wrapper + interruption monitor	1x GPU	<400 ms p50 demo
5	36-44	DPO-lite preference optimization	1x GPU	Human A/B beats SFT
6	45-52	Voice/persona conditioning and consent flow	1x GPU	3 personas stable
7	53-60	Latency polish, eval, launch page	1x GPU	Public demo

8.2 Hyperparameter starting point

LoRA rank: 32 first, 64 if stable
 LR: 1e-5 to 3e-5 for SFT, 1e-7 to 5e-7 for DPO-lite
 Batch: maximize tokens/audio frames; use gradient accumulation
 Precision: bf16 where possible; 4-bit QLoRA if VRAM constrained
 Freeze: codec decoder and most backbone blocks at first
 Unfreeze: embeddings for new emotion tags, adapter layers, final projection
 Loss: token NLL + optional emotion consistency auxiliary after baseline works

8.3 Do not do in 60 days

- Do not pretrain a codec from scratch.
- Do not scale to 8B from scratch.
- Do not build a custom CUDA kernel before a working demo exists.
- Do not launch arbitrary voice cloning without consent gates.

9. Latency plan - how to hit the lowest practical delay

9.1 Latency budget

Component	Day-30 target	Day-60 target	Tactics
Mic/WebRTC/VAD	60 ms	40 ms	Small chunks, no large jitter buffer
Speech/dialog model	200 ms	120 ms	Warm KV cache, small context, prefill persona
TTS/codec decode	180 ms	90 ms	Chunked decode, CUDA graphs, compiled decoder
Network/playback	80 ms	50 ms	Same-region hosting, browser audio buffers tuned
Total	<520 ms	<300 ms	Measure p50/p95 on real calls

9.2 Optimization order

- First remove architecture hops. Speech-to-speech or streaming TTS beats batch TTS.
- Second warm everything: model weights, KV cache, tokenizer, first persona prompt.
- Third reduce chunk sizes and push audio before the whole sentence is ready.
- Fourth use quantization and torch.compile/CUDA graphs.
- Only after all of that, write custom Triton/CUDA kernels.

9.3 Why vLLM concepts matter

PagedAttention/vLLM improves serving throughput by handling KV cache memory more efficiently and enabling high-throughput batched serving [R4]. For a solo 60-day build, you may not need to fork vLLM; you do need to copy the mindset: keep cache warm, avoid memory fragmentation, and avoid repeated prompt prefill.

10. Day-by-day 60-day execution calendar

Week 1

Day	Work
Day 1	Rent GPU, create repo, download base weights, run baseline demo.
Day 2	Measure baseline TTFB/WER/MOS. Create eval harness.
Day 3	Create data schema: transcript, speaker, emotion tags, consent metadata.
Day 4	Implement synthetic Hinglish text generator and first 1k lines.
Day 5	Record first 200 real emotional utterances.
Day 6	Build audio cleaning + back-transcription filters.
Day 7	Freeze baseline report: what works, what breaks, exact latency.

Week 2

Day	Work
Day 8	Add emotion tag vocabulary and tokenizer handling.
Day 9	Generate 5 hr synthetic emotional Hinglish audio.
Day 10	Run WER/emotion filters; create train/val/test splits.
Day 11	Implement LoRA training script.
Day 12	Run smoke fine-tune on 30 min data.
Day 13	Debug artifacts: pronunciation, laughs, pauses, silence.
Day 14	Run v1 fine-tune overnight.

Week 3

Day	Work
Day 15	Evaluate v1, collect 100 human notes from your own listening.

Day	Work
Day 16	Scale dataset to 10-15 hr and rebalance emotions.
Day 17	Add persona prompt templates.
Day 18	Train v2.
Day 19	Build web demo skeleton with mic-in/audio-out.
Day 20	Integrate streaming output chunks.
Day 21	Rest + only monitor runs; do not start new architecture.

Week 4

Day	Work
Day 22	Implement interruption monitor with VAD.
Day 23	Tune stop/fade behavior on user interruption.
Day 24	Create three personas with fixed voices.
Day 25	Record additional real data for weakest persona.
Day 26	Train v3 with persona balanced data.
Day 27	Latency pass: warm cache, chunk sizes, no batch stalls.
Day 28	Private demo to 10 friends; collect A/B ratings.

Week 5

Day	Work
Day 29	Generate preference pairs: v2 vs v3 vs baseline.
Day 30	Label 500-1000 pairs manually/model-assisted.
Day 31	Run DPO-lite experiment.
Day 32	Evaluate DPO for naturalness regressions.
Day 33	Scale to 2k-5k preference pairs.
Day 34	Train v4.
Day 35	Create comparison clips and blind survey.

Week 6

Day	Work
Day 36	Consent-only voice reference flow.
Day 37	Speaker embedding/ref prompt integration.
Day 38	Safety block for celebrity/public-figure cloning requests.
Day 39	Watermarking/provenance prototype.
Day 40	Stress test noisy rooms and interruption.
Day 41	Train v5 with final data mix.
Day 42	Rest + freeze feature list.

Week 7

Day	Work
Day 43	Productionize server: process supervision, logs, metrics.
Day 44	GPU memory optimization and quantization test.
Day 45	p50/p95 latency load test.
Day 46	Eval suite: WER, MOS proxy, emotion classifier, human A/B.
Day 47	Fix top 5 failure modes only.
Day 48	Build landing page + waitlist + demo gating.
Day 49	Record polished demo videos.

Week 8

Day	Work
Day 50	Security review: consent, rate limits, prompt abuse.
Day 51	Final train or rollback decision.
Day 52	Write model card, data card, safety card.
Day 53	Deploy staging.
Day 54	20-user private beta.
Day 55	Patch only launch blockers.
Day 56	Public demo launch.

Buffer

Day	Work
Day 57	Critical bug fixes from launch.
Day 58	Additional A/B eval and investor-ready metrics.
Day 59	Clean repo, docs, and reproducible scripts.
Day 60	Publish technical write-up and next 90-day roadmap.

11. Claude Code + GPT Codex orchestration

Use both agents like a two-engineering-team setup: Claude for architecture/refactor/reasoning, Codex for isolated implementation, tests, kernels, and scripts. Always use git worktrees so the agents cannot overwrite each other.

```
git worktree add ../vox-data data-pipeline
git worktree add ../vox-stream streaming
git worktree add ../vox-train training

# Claude Code in ../vox-train:
# Read docs/specs/sft.md and implement training/sft.py with tests.

# Codex in ../vox-data:
# Implement data/pipelines/synthesize_codeswitch.py and unit tests.
```

Agent guardrails

- No agent edits production configs without a human diff review.
- Every agent task ends with tests and a commit.
- Do not allow agents to launch large GPU jobs unsupervised.
- One markdown spec per feature before coding.

12. Repo skeleton + exact first commands

```
vox-omega/
  configs/
    base.yaml
    emotion_tokens.yaml
    training_lora.yaml
    inference.yaml
  data/
    pipelines/clean_audio.py
    pipelines/synthesize_codeswitch.py
    pipelines/mimi_tokenize.py
    manifests.py
  model/
    emotion_tags.py
    speaker_conditioning.py
    moshi_adapter.py
  training/
    sft.py
    dpo_lite.py
    eval_callback.py
  inference/
    server.py
    webrtc_transport.py
    interrupt_monitor.py
```

```
latency_probe.py
eval/
  wer.py
  emotion_acc.py
  human_ab.py
  latency.py
safety/
  consent.py
  watermark.py
  abuse_filters.py
docs/
  model_card.md
  data_card.md
  safety_card.md

# Day 1 commands
mkdir -p vox-omega && cd vox-omega
git init
python3.11 -m venv .venv
source .venv/bin/activate
pip install --upgrade pip
pip install torch torchaudio transformers accelerate peft trl datasets soundfile librosa fastapi uvicorn wandb hydra-co
pip install livekit pipecat-ai || true
huggingface-cli login
wandb login

# First checkpoint: run any chosen base voice model locally and measure baseline latency.
```

13. Evaluation gates and launch scorecard

Metric	Tool	Pass threshold	Launch threshold
First audible response	latency.py	<500 ms p50	<300 ms p50
Interruption latency	interrupt_eval.py	<350 ms	<250 ms
Intelligibility	Whisper back-transcribe	WER <15%	WER <10%
Emotion accuracy	Classifier + human	>60%	>75%
Naturalness	Blind human A/B	>50% vs baseline	>60% vs baseline
Hinglish fluency	Human raters	3.5/5	4.2/5
Safety	Red-team prompts	No arbitrary clone	Consent + watermark + logs

Do not claim 'beats Hume' until a blind A/B panel says so on a fixed set. You can claim: 'our public demo is optimized for faster, more expressive Hinglish emotional voice than generic voice agents.'

14. Safety, consent, watermarking, legal guardrails

Voice cloning is high-risk because it can enable impersonation, scams, and non-consensual deepfakes. The product should ship with safety from the first prototype, not after launch.

Risk	Guardrail
Unconsented voice cloning	Require consent phrase and account verification before cloning.
Celebrity/public figure impersonation	Block requests to clone or imitate identifiable public figures.
Fraud/scams	Rate limit, watermark, abuse detection, no phone-call automation during MVP.
Dataset rights	Only use licensed/open/consented audio. Keep a data ledger.
Synthetic content disclosure	Add watermark/provenance and clear UI disclosure.
Children/minors	No cloning minors; strict age gating for public demo.

AudioSeal-style watermarking is directly relevant because it embeds inaudible signals for detecting generated audio and is intended for localized detection in longer clips [R7].

15. Budgets: solo, serious, and funded

Path	Timeline	Team	Compute/data budget	Expected outcome
Solo vacation MVP	60 days	1 founder + AI coding agents	\$2k-\$8k	Demo with 3 personas, Hinglish emotion, <300-500 ms
Serious seed prototype	3-4 months	3-4 engineers	\$50k-\$200k	Reliable beta, real data, stronger DPO, lower latency

Path	Timeline	Team	Compute/data budget	Expected outcome
Frontier competitor	6-12 months	7-12 people	\$500k-\$2M+	Custom codec/backbone, robust multilingual eval, commercial readiness

Best use of the next \$5,000

- GPU rental for continuous experiments.
- Human voice actors for consented emotional Hinglish data.
- Human A/B labels for preference optimization.
- A clean landing page and 20-user private beta, not vanity benchmarks.

16. Bibliography and source index

ID	Source
R1	Defossez et al. 'Moshi: a speech-text foundation model for real-time dialogue.' arXiv:2410.00037. https://arxiv.org/abs/2410.00037
R2	Yang et al. 'EmoVoice: LLM-based Emotional Text-To-Speech Model with Freestyle Text Prompting.' arXiv:2504.12867. https://arxiv.org/abs/2504.12867
R3	Qin et al. 'OpenVoice: Versatile Instant Voice Cloning.' arXiv:2312.01479. https://arxiv.org/abs/2312.01479
R4	Kwon et al. 'Efficient Memory Management for Large Language Model Serving with PagedAttention.' arXiv:2309.06180. https://arxiv.org/abs/2309.06180
R5	Tom's Guide, Hume EVI 3 custom voices and safeguards reporting, 2025.
R6	WIRED, Google DeepMind/Hume AI talent and licensing reporting, 2026.
R7	Meta AudioSeal reporting and release context, 2024.
R8	Nishimura et al. 'HALL-E: Hierarchical Neural Codec Language Model for Minute-Long Zero-Shot Text-to-Speech Synthesis.' arXiv:2410.04380.
R9	Singh et al. 'Sample-Efficient Language Model for Hinglish Conversational AI.' arXiv:2504.19070.
R10	Hutiri et al. 'Not My Voice! A Taxonomy of Ethical and Safety Harms of Speech Generators.' arXiv:2402.01708.
U1	User-provided VOX_OMEGA_MASTER_PLAN.md and VOX_OMEGA_BUNDLE.zip.
U2	User-provided Vatsal Bharti resume screenshot.

Important limitation: some official pages were not retrievable in the browsing environment. Where an official page could not be fetched, this plan relies on accessible papers, news reporting, and the user-provided uploaded materials. Any investor-facing claim should be rechecked against the original official pages before publication.